



OPERATIONAL MANUAL

DOOR INTERLOCKING SYSTEM – 2 DOOR

MODEL NO: DLS153-2D

VERSION: 1.0



NEURAL INDUSTRIAL AUTOMATION PVT LTD.,

PLOT NO: 16 Survey No: 60, Shanthi Nagar,
Near Aparan Palm Meadows,
Kompally, Hyderabad – 500 100
Telangana, INDIA

Ph: +91 79958 65752

www.neuralindustrialautomation.com

Email: [sales@ neuralindustrialautomation.in](mailto:sales@neuralindustrialautomation.in)

Dear Customer,

Thank you for choosing a NEURAL instrument, please read this operating manual carefully before you start using the instrument. This manual will provide you with all the necessary information for the correct use of the instrument as well as a precise idea of its versatility in a wide range of applications.

CAUTION

One three way connector is provided for power connection i.e. Line, Neutral & Earth. (100-220V AC 50/60Hz.)



Do not Switch ON power to the instrument before plugging connectors in to the terminals.



Carefully connect the connectors in the specified terminals if not it may cause damage for the instrument.

Instrument SI. No :

STANDARD PACK

- | | | | |
|-------------------------|-----|------|--------------------------|
| 1) Instruction Manual | --- | 1 No | ☐ |
| 2) DLS153-2D Instrument | --- | 1 No | ☐ |

Checked By :

Date :

1. DESCRIPTION:

In clean room areas, any personnel passing through an unclassified area to a classified area is required to pass through series of rooms. In each room certain steps need to be followed. This procedure ensures that the person is not carrying any external contamination in to the process area and also the pressure in each room is maintained before opening of door.

Inter Locking System achieves in prevention of process area contamination by

- Preventing simultaneous access of doors/Rooms.
- Ensures that when one door for a room is OPEN then other door in that particular room is closed.

2. Controller Features:

- ✓ Micro controller based.
- ✓ Supports Maximum of 2 doors.
- ✓ It can support Emergency Stations.
- ✓ Delay time between door openings is 4 Sec.
- ✓ Inbuilt sensor Magnetic Lock to read door feed back.
- ✓ Relay status indication
- ✓ Door status indication

3. Controller Specifications:

GENERAL

- Push Button Inputs : 2 No.
- Door Feedback Inputs : 2 No, 24V.
- LED indication Outputs for push buttons : 2 No.
- Status LED : GO, WAIT Indication or Red and Green
- Door control outputs : 2No, 24V DC. load current

RELAYS:

- Contact material : Silver Alloy.
- Contact Rating : 230V AC / 5 A.
- No. of Operations : Mechanical : 10^7 , Electrical : 10^5 .

ENCLOSURE:

- Enclosure type : MS Powder coated
- Mounting : Wall mounting

POWER:

- Power source : 100 – 280V AC 50 Hz
- Wattage : 30W

ENVIRONMENTAL:

- Operating Temperature : 0 to 50°C.

MECHANICAL:

- Weight : 1.5 Kgs.
- Dimensions : 180X240X50mm

4. Push Button Station

Features:

- ✓ Elegant look North-west SS Face plate.
- ✓ Indication facility for Door entry permitted / prohibited.
- ✓ Support an entry switch for opening the door.
- ✓ Long Life BAR LED for door open/close indication.
- ✓ Elegant look North-west SS Face plate with appropriate curvature to prevent resting of dust & powder.
- ✓ Easy mounting arrangement for fixing on Doorframes or on Wall.



Specifications:

- | | | |
|-----------------------|---|--|
| ➤ Keypad | : | GENERAL:
Soft touch Tactile Membrane Key |
| ➤ Status LED's | : | WAIT Message for Door Open / Entry Prohibited.
GO Message for Door Close / Entry Permitted. |
| ➤ Digital Input (LED) | : | 24V DC |
| ➤ Life | : | 100, 000 Cycles Min. |

ENCLOSURE:

- | | | |
|------------------|---|---------------------------|
| ➤ Enclosure type | : | North-west SS Face plate. |
| ➤ Mounting | : | Flush mounting. |
| ➤ Dimension | : | 3"X3" Single Module. |

POWER:

Supply Voltage : 24V/50mA DC

5. Emergency Station**Features:**

- ✓ Elegant look North-west SS Face plate.
- ✓ Support an push button switch for opening the all doors.
- ✓ Elegant look North-west SS Face plate with appropriate curvature to prevent resting of dust & powder.
- ✓ Easy mounting arrangement for fixing on Doorframes or on Wall.

**Specifications:**

- | | | |
|-----------------------|---|--|
| ➤ Keypad | : | GENERAL:
Soft touch Tactile Membrane Key |
| ➤ Digital Input (LED) | : | 24V DC |
| ➤ Life | : | 100, 000 Cycles Min. |

ENCLOSURE:

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| ➤ Enclosure type | : | North-west SS Face plate. |
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| ➤ Dimension | : | 3"X3" Single Module. |

POWER:

Supply Voltage : 24V/50mA DC

6. Electromagnetic locks:

A **magnetic lock** is a simple locking device that consists of an electromagnet and armature plate. By attaching the electromagnet to the door frame and the armature plate to the door, a current passing through the electromagnet attracts the armature plate holding the door shut.

The magnetic lock relies upon some of the basic concepts of electromagnetism. Essentially it consists of an electromagnet attracting a conductor with a force sufficiently large enough to prevent the door from being opened.



7. Description of operation modes:

Model1:

Normally Close:

In this mode, initially both doors are in a locked condition. Door access is by using a push button. Message indication provided on the push button station indicates whether a person can access the door or not.

Go = Indicates that a person can access the door.

Wait = Indicates that a person cannot access the door.

When the door is operated from open to closed condition, both doors are locked for a 4-second door delay time, during which a person cannot access both doors. If an emergency button is pressed, both doors will be unlocked up to 8 seconds and message indication on the push button will be toggled.

Mode 2:

Normally Open:

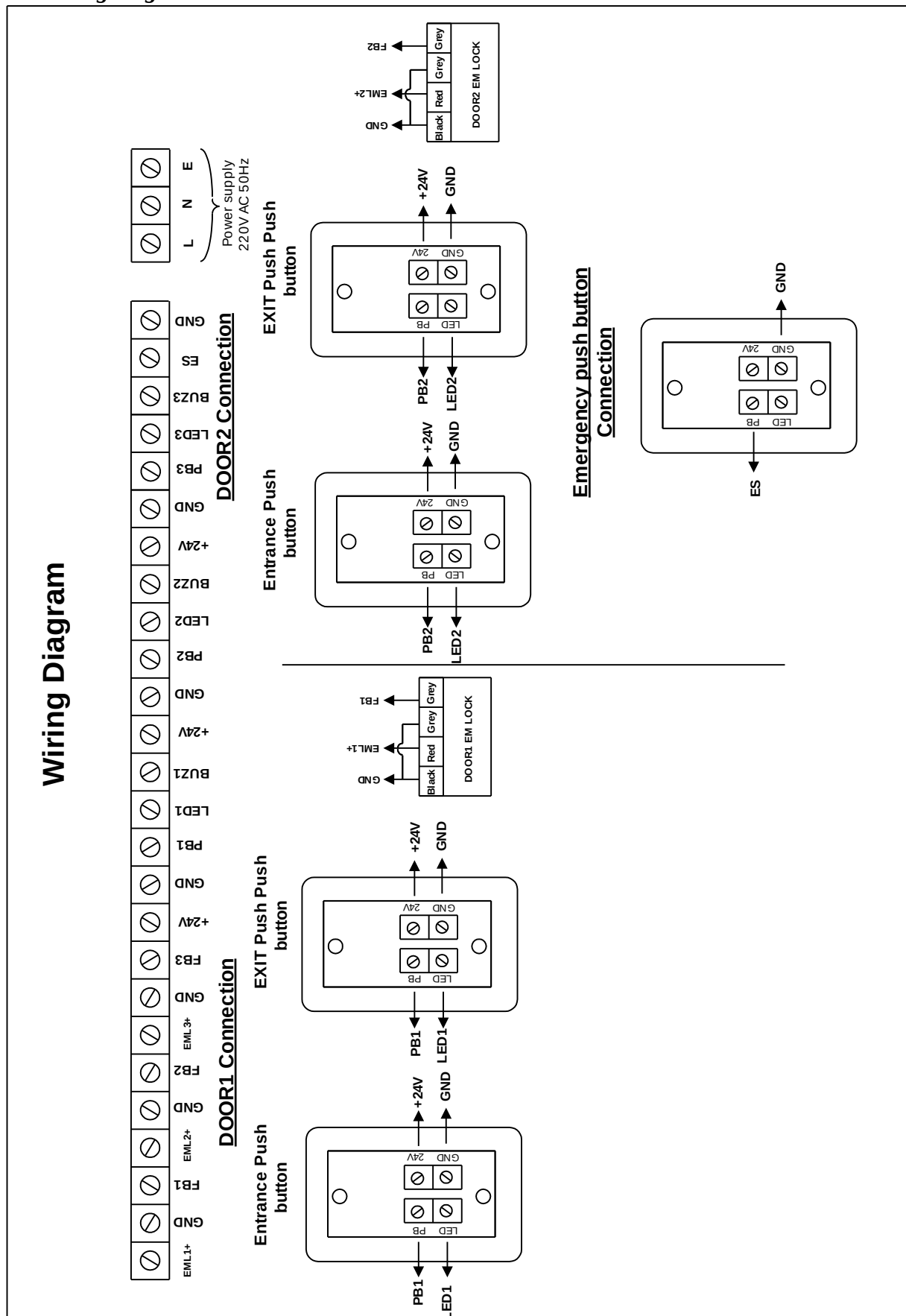
In this mode, initially both doors are in an unlocked condition; here push buttons are used only for indication purposes (no need to press the button). Message indication provided on the push button station indicates whether a person can access the door or not.

Go = Indicates that a person can access the door.

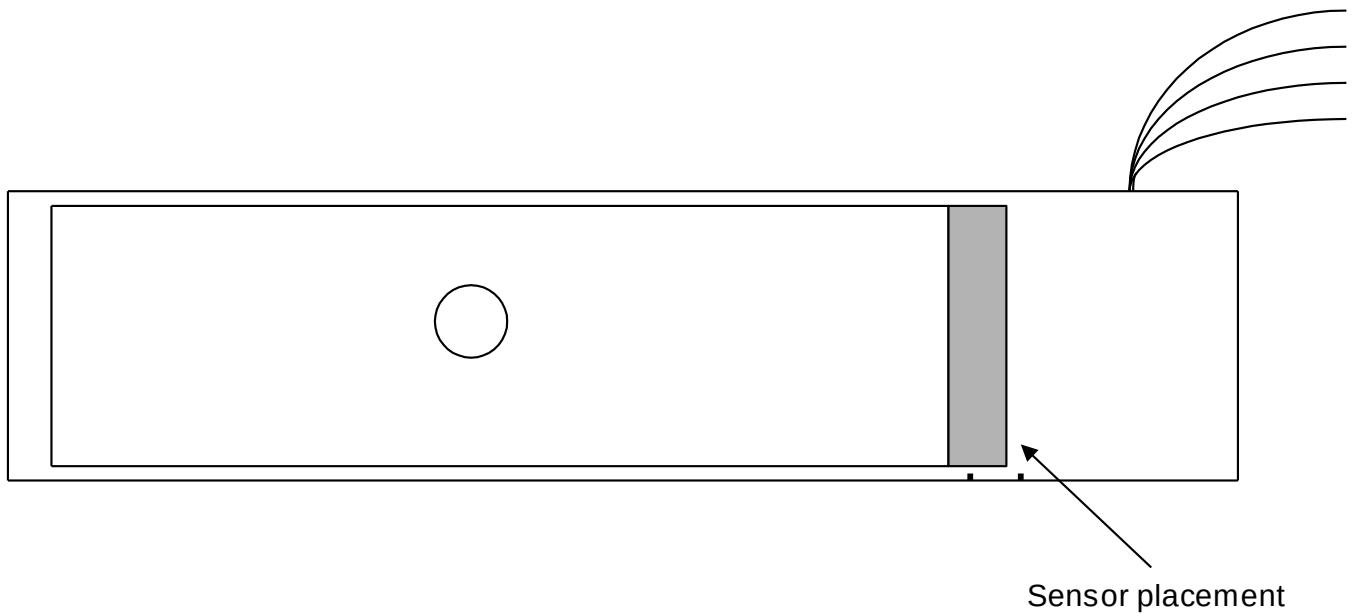
Wait = Indicates that a person cannot access the door.

When the door is operated from open to closed condition, both doors are locked for a 4-second door delay time, during which a person cannot access both doors. If an emergency button is pressed, both doors will be unlocked up to 8 seconds and message indication on the push button will be toggled.

8. Wiring diagram:

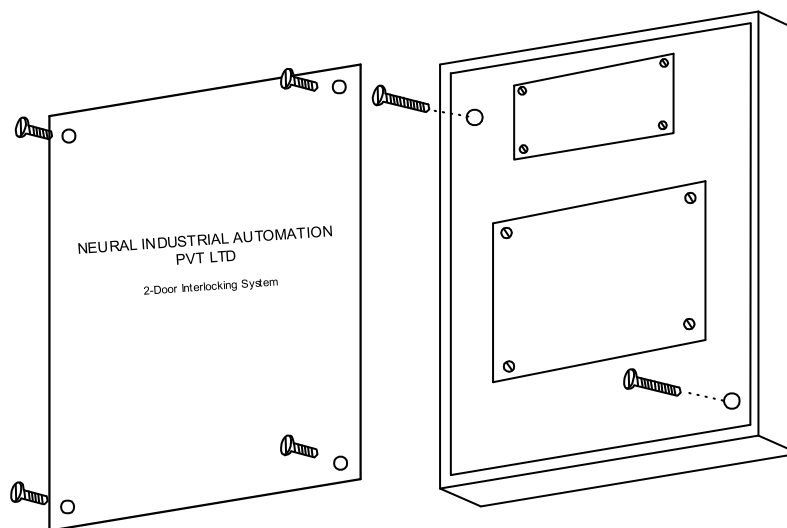


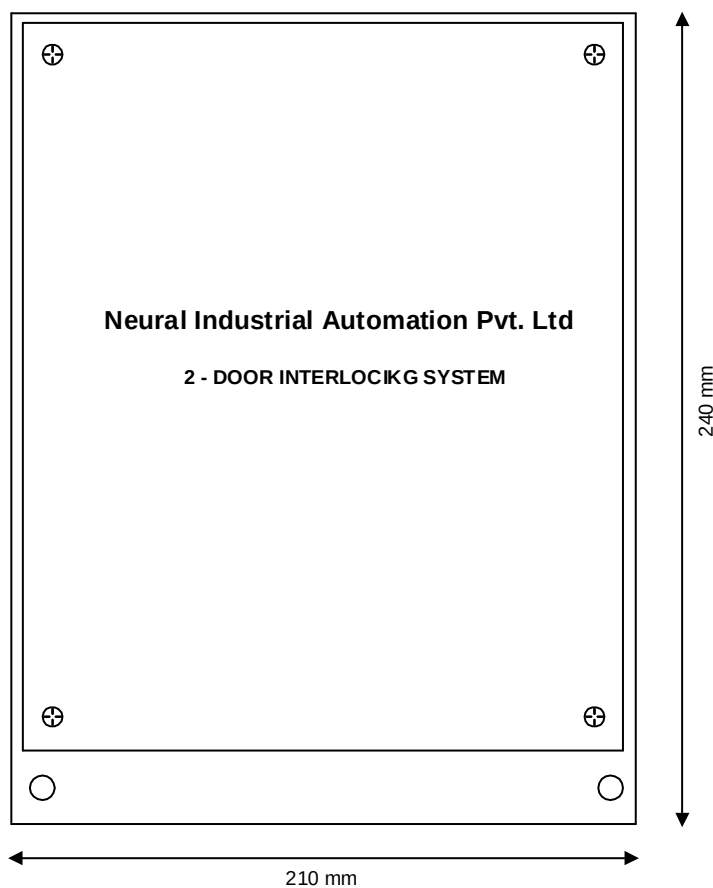
9. Magnetic lock sensor placement:



10. MECHANICAL DRAWINGS

Figure for Installation of DLS153-2D Controller:



Front View Diagram of DLS153-2D Controller:**11. ORDERING INFORMATION****1). Door Interlocking System (2 Door):**

<i>Model No</i>	<i>X</i>
DLS153-2D	X

includes,

b) Door interlock controller(2Door):

<i>Model No</i>	<i>X</i>
DLS153-2C	X

b) Push Button Station:

<i>Model No</i>	<i>X</i>
DLS153-P	NORTH WEST PUSH BUTTON
DLS153-NP	NO-TOUCH SWITCH

c) Emergency Station:

<i>Model No</i>	<i>X</i>
DLS153-E	NORTH WEST PUSH BUTTON

d) Electromagnetic Lock:

<i>Model No</i>	<i>X</i>
EML-B	1. Bar Electro magnetic lock

12. TROUBLE SHOOTING

SL.NO	Problem	Cause	Action
1.	No Message indication.	Power Failed	Check the Power supply connections.
2.	Indication on Push button blinking.	Noise	Switch OFF and switch ON again. If problem is reported again then contact to the factory.
3.	Continuous WAIT indication on push button	Other door is not closed properly or magnet armature not fixed properly	Adjust armature until get GO

NOTE:

For Further Information Contact:

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